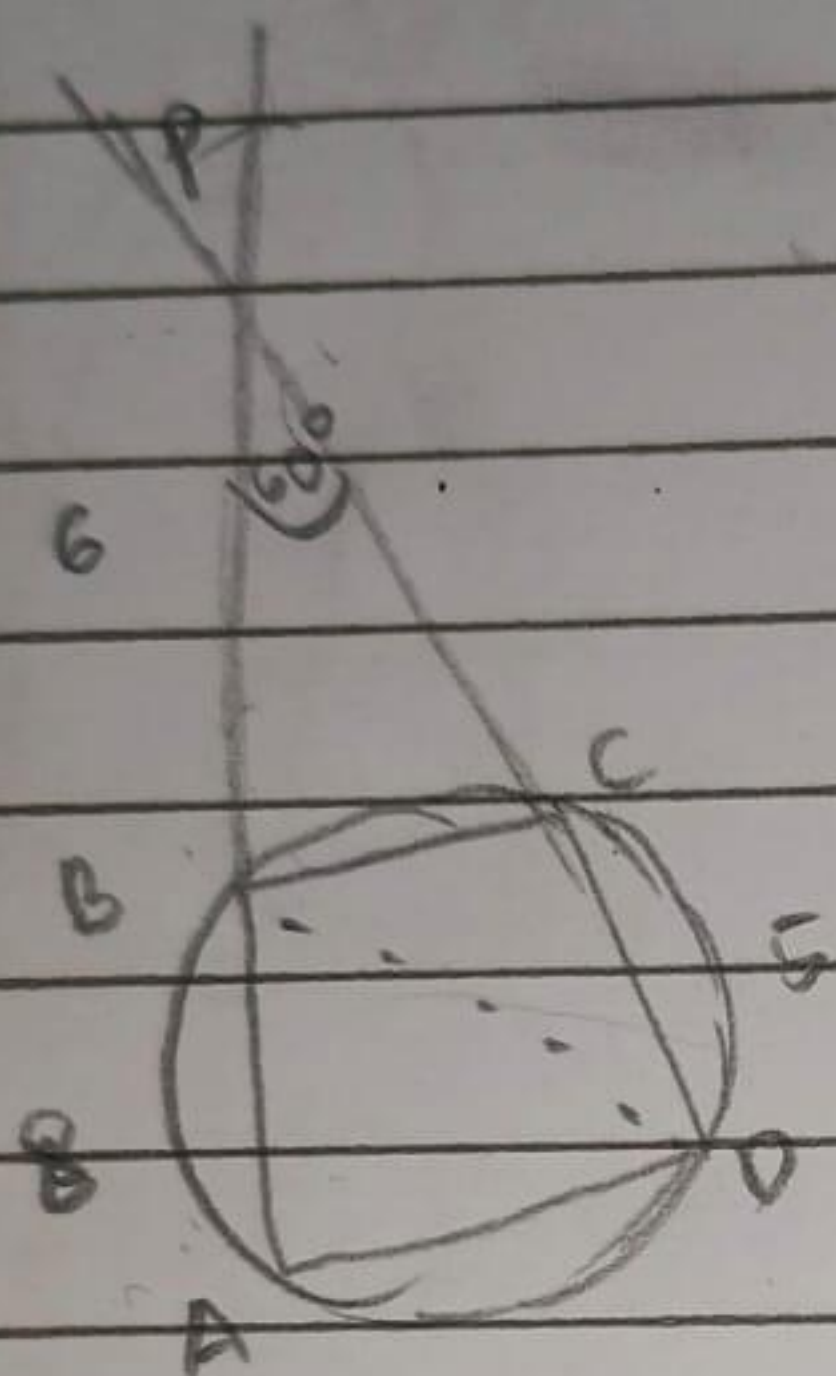


Pada suatu lingkaran dgn jml = ...
 talibusur ABCP dgn $AB = 8$ dan $CD = 5$ - sisi AB dan
 DC diperpanjang dan berpotongan di luar lingkaran di titik P.
 $\angle APD = 60^\circ$ dan $BP = 6$, maka nilai r^2 adalah...

Jawab:



$$1) PA \cdot PB = PC \cdot PD$$

$$(6+8) \cdot 6 = x \cdot (PC + CD)$$

$$14 \cdot 6 = x(x+5)$$

$$84 = x^2 + 5x$$

$$x^2 + 5x - 84 = 0$$

$$(x+12)(x-7) = 0$$

$$x = -12 \vee x = 7$$

Jadi $PC = 7$ (panjang harus +) $\Rightarrow PD = 7 + 5 = 12$

$$2) BC^2 = PB^2 + PC^2 - 2 \cdot PB \cdot PC \cdot \cos 60^\circ$$

$$BC^2 = 6^2 + 7^2 - 2 \cdot 6 \cdot 7 \cdot \frac{1}{2}$$

$$BC^2 = 36 + 49 - 42$$

$$BC^2 = 43$$

$$BC = \sqrt{43}$$

$$BD^2 = PD^2 + PB^2 - 2 \cdot PD \cdot PB \cdot \cos 60^\circ$$

$$BD^2 = 12^2 + 6^2 - 2 \cdot 12 \cdot 6 \cdot \frac{1}{2}$$

$$BD^2 = 144 + 36 - 72$$

$$BD^2 = 108$$

$$BD = \sqrt{108} = 6\sqrt{3}$$

$$3) BC^2 = BD^2 + CD^2 - 2 \cdot BD \cdot CD \cdot \cos D$$

$$43 = 108 + 5^2 - 2 \cdot 6\sqrt{3} \cdot 5 \cdot \cos D$$

$$43 = 133 - 60\sqrt{3} \cos D$$

$$60\sqrt{3} \cos D = 90$$

$$\cos D = \frac{90}{60\sqrt{3}} = \frac{3}{2\sqrt{3}} = \frac{\sqrt{3}}{2} \quad \angle D = 30^\circ$$

$$\text{Jadi, } \sin 30^\circ = \frac{1}{2}$$

4)

BC

$r =$

$$2 \cdot \sin \angle BDC$$

$$\sqrt{43}$$

$r =$

$$2 \cdot \sin 30^\circ$$

$$\sqrt{43}$$

$r =$

$$2 \cdot \frac{1}{2}$$

$$r^2 = 43$$

